DRAFT PROVISIONAL PATENT APPLICATIONS

Series B: six emergent combinations joining JoinAQAL's implemented integration API to Russell Capital Systems' implemented Decade Machine modules. Both sides run; the joint orchestration is specified.

STATUS: DRAFT — NOT FILED. No application numbers exist; nothing herein is patent-pending. Per company records; confirm status with patent counsel. Prepared with AI assistance; not legal advice; attorney review required before filing.

Inventor: Samuel Andrew Russell V — Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E1

Title of Invention: Voice-Informed Financial Readiness Assessment

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine; joined across platforms with the Decade Machine household-profile intake and module-triage layer (shared/ultraEngine.ts; server/ultraAI.ts). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. A member's prosodic voice profile and consensus-scored cognitive vector, retrieved over the authenticated integration API, condition the financial planning engine's module triage: the same spoken self-description that drives cognitive assessment also seeds the household profile, and voice-derived confidence context is disclosed alongside the plan rather than hidden inside it.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Eight-Model Consensus Scoring + Transparency and Tamper-Evident Audit across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight w = 1/(meanAbsErr + 0.02) with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = 0.5 x mean model confidence + 0.5 x inter-model agreement.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a multi-platform computing system by producing a tamper-evident emergent cross-platform result, the method comprising: retrieving, over an authenticated, versioned application programming interface in which every response is scoped to the authenticated member and authentication failures are rejected fail-closed, member data produced by: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) providing the retrieved data to the Decade Machine household-profile intake and module-triage layer (shared/ultraEngine.ts; server/ultraAI.ts); (e) combining the platforms' outputs into a single emergent result not produced by either platform alone; and (f) committing the emergent result, with a disclosure of every input, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.

6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame

length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **9.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.5 / 10**

- Basis for reduction-to-practice score: Both platform sides implemented and running; the joint orchestration is specified but not yet deployed as one system.

- **Path to 9.7:** Deploy the specified cross-platform orchestration against the live /api/v1/joinaqal/* surface and record one end-to-end run to the ledger — that single action lifts reduction to practice to 10.0 and overall readiness to 9.7.

Nomenclature Note

NOMENCLATURE NOTE. Earlier planning documents prepared by the collaborating AI team enumerated emergent and super-emergent combinations under other working titles. Those documents were exchanged in chat and are not preserved verbatim in this repository; the combinations below are drafted directly from the implemented and specified systems themselves. Counsel should reconcile titles against any earlier enumerations before filing.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E2

Title of Invention: Weakness-Aware Financial Coaching

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine; joined across platforms with the RCS weakness-interventions and planning-case modules (server/patents/weaknessInterventions.ts in the cognitive domain; RCS planning cases and coaching surfaces in the financial domain). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. The controlling weakness identified by the cognitive argmin routes the member to paired interventions: the evidence-mapped development protocol for the weak line AND the financial-planning module whose relevance rule matches that weakness, so cognitive and financial coaching are issued as one transparent, ledger-committed plan.

Abstract

A computer-implemented system combining Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.

- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a multi-platform computing system by producing a tamper-evident emergent cross-platform result, the method comprising: retrieving, over an authenticated, versioned application programming interface in which every response is scoped to the authenticated member and authentication failures are rejected fail-closed, member data produced by: (a) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) providing the retrieved data to the RCS weakness-interventions and planning-case modules (server/patents/weaknessInterventions.ts in the cognitive domain; RCS planning cases and coaching surfaces in the financial domain); (e) combining the platforms' outputs into a single emergent result not produced by either platform alone; and (f) committing the emergent result, with a disclosure of every input, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **9.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.5 / 10**

- Basis for reduction-to-practice score: Both platform sides implemented and running; the joint orchestration is specified but not yet deployed as one system.

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Disclaimer

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E3

Title of Invention: Cross-Domain Development Tracker

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine; joined across platforms with the Decade Machine's chained planning windows (every window inherits the previous window's ending state). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. Cognitive development milestones (floor raises, weakness migration, stage-band movement) recorded on the JoinAQAL side are carried into the financial engine's next planning window as changed capacity assumptions, so the financial projection and the cognitive development plan advance on one shared timeline.

Abstract

A computer-implemented system combining Developmental Stage-Band + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.

- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.

- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a multi-platform computing system by producing a tamper-evident emergent cross-platform result, the method comprising: retrieving, over an authenticated, versioned application programming interface in which every response is scoped to the authenticated member and authentication failures are rejected fail-closed, member data produced by: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome

is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) providing the retrieved data to the Decade Machine's chained planning windows (every window inherits the previous window's ending state); (e) combining the platforms' outputs into a single emergent result not produced by either platform alone; and (f) committing the emergent result, with a disclosure of every input, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **9.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.5 / 10**

- Basis for reduction-to-practice score: Both platform sides implemented and running; the joint orchestration is specified but not yet deployed as one system.

- **Path to 9.7:** Deploy the specified cross-platform orchestration against the live /api/v1/joinaqal/* surface and record one end-to-end run to the ledger — that single action lifts reduction to practice to 10.0 and overall readiness to 9.7.

Nomenclature Note

NOMENCLATURE NOTE. Earlier planning documents prepared by the collaborating AI team enumerated emergent and super-emergent combinations under other working titles. Those documents were exchanged in chat and are not preserved verbatim in this repository; the combinations below are drafted directly from the implemented and specified systems themselves. Counsel should reconcile titles against any earlier enumerations before filing.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E4

Title of Invention: Transparent Cross-Domain Planning Ledger

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Transparency and Tamper-Evident Audit Engine; joined across platforms with the Decade Machine's projection disclosure layer (every output labeled as a projection under stated assumptions). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. Every cross-domain emergent result — cognitive score, plan triage, projection window — is committed to the same append-only hash chain, giving the member one verifiable audit trail across both platforms; chain validity is publicly checkable and the chain head is exportable for external anchoring.

Abstract

A computer-implemented system combining Transparency and Tamper-Evident Audit across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a multi-platform computing system by producing a tamper-evident emergent cross-platform result, the method comprising: retrieving, over an authenticated, versioned application programming interface in which every response is scoped to the authenticated member and authentication failures are rejected fail-closed, member data produced by: (a) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (b) providing the retrieved data to the Decade Machine's projection disclosure layer (every output labeled as a projection under stated assumptions); (c) combining the platforms' outputs into a single emergent result not produced by either platform alone; and (d) committing the emergent result, with a disclosure of every input, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

5. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

6. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

7. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a

confidence threshold, and floors only rise.

8. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **9.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.5 / 10**

- Basis for reduction-to-practice score: Both platform sides implemented and running; the joint orchestration is specified but not yet deployed as one system.

- **Path to 9.7:** Deploy the specified cross-platform orchestration against the live /api/v1/joinaqal/* surface and record one end-to-end run to the ledger — that single action lifts reduction to practice to 10.0 and overall readiness to 9.7.

Nomenclature Note

NOMENCLATURE NOTE. Earlier planning documents prepared by the collaborating AI team enumerated emergent and super-emergent combinations under other working titles. Those documents were exchanged in chat and are not preserved verbatim in this repository; the combinations below are drafted directly from the implemented and specified systems themselves. Counsel should reconcile titles against any earlier enumerations before filing.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E5

Title of Invention: Consensus-Scored Wealth-Capacity Profile

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine; joined across platforms with the RCS multi-factor client-fit profile (the 25-factor financial-fit intake of the RCS prototype family). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. The eight-model consensus vector supplies the cognitive-capacity factors of the financial-fit profile, replacing self-report on those factors with calibration-weighted panel measurement, and the combined profile discloses which factors are measured versus self-reported.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Transparency and Tamper-Evident Audit across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight w = 1/(meanAbsErr + 0.02) with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = 0.5 x mean model confidence + 0.5 x inter-model agreement.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a multi-platform computing system by producing a tamper-evident emergent cross-platform result, the method comprising: retrieving, over an authenticated, versioned application programming interface in which every response is scoped to the authenticated member and authentication failures are rejected fail-closed, member data produced by: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each

entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) providing the retrieved data to the RCS multi-factor client-fit profile (the 25-factor financial-fit intake of the RCS prototype family); (d) combining the platforms' outputs into a single emergent result not produced by either platform alone; and (e) committing the emergent result, with a disclosure of every input, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.

6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **9.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.5 / 10**

- Basis for reduction-to-practice score: Both platform sides implemented and running; the joint orchestration is specified but not yet deployed as one system.

- **Path to 9.7:** Deploy the specified cross-platform orchestration against the live /api/v1/joinaqal/* surface and record one end-to-end run to the ledger — that single action lifts reduction to practice to 10.0 and overall readiness to 9.7.

Nomenclature Note

NOMENCLATURE NOTE. Earlier planning documents prepared by the collaborating AI team enumerated emergent and super-emergent combinations under other working titles. Those documents were exchanged in chat and are not preserved verbatim in this repository; the combinations below are drafted directly from the implemented and specified systems themselves. Counsel should reconcile titles against any earlier enumerations before filing.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-RCS-E6

Title of Invention: Floor-Verified Advisor-Client Matching

Series: Series B — JoinAQAL x Russell Capital Systems Emergent Combination

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; cross-platform financial-planning integration; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it; and cognitive and financial development systems operate as silos, so what one system learns about a person never conditions the other.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Controlling-Weakness Identification Engine, Transparency and Tamper-Evident Audit Engine; joined across platforms with the RCS constraint-satisfaction team-formation engine (server/patents/teamFormation.ts: no two members share a controlling weakness; per-axis coverage maximized). The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Emergent mechanism. Advisor-client and team pairings are computed over VERIFIED floors (repeated-demonstration minimums) rather than single-session scores, and under the constraint that no assembled team doubles its weakest link, with every served match committed to the ledger.

Abstract

A computer-implemented system combining Controlling-Weakness Identification + Transparency and Tamper-Evident Audit across authenticated platform boundaries to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Cross-platform data path. Member data crosses platforms only over the authenticated integration API (/api/v1/joinaqal/*): eight versioned REST endpoints, session-authenticated, fail-closed on authentication errors, every response scoped to the authenticated member. *Implementation:* server/integrationApi.ts (JoinAQAL side, operating).

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). This is a CROSS-COMPANY combination. The JoinAQAL surface is IMPLEMENTED AND OPERATING: eight versioned REST endpoints under /api/v1/joinaqal/* (assessment create/read, voice analysis, development tracking, weakness identification, coaching create/read, transparency), each authenticated fail-closed and scoped to the authenticated member, per the API Integration Specification v1.0. The Russell Capital Systems components named below are IMPLEMENTED in the separate RCS codebase (the Decade Machine engine and its modules). The cross-company ORCHESTRATION that joins the two running systems into one deployed product is SPECIFIED but not yet deployed as a single operating system, and no filing should claim that the combined system runs in production today.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

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3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

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